



VPC Traffic Flow and Security



Thato Ntoahae

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User name

[nextwork-dev-thato](#)

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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is... a virtual private cloud and it is useful because...it offers full control over IP ranges, subnets, and routing, enhancing security and scalability for cloud resources.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to...create route tables and subnets also amazon ACLs

One thing I didn't expect in this project was...

One thing I didn't expect in this project was...creating an ACL because i didnt think my vpc needed a second layer of defence.

This project took me...

This project took me...about 3 hours to complete.

Route tables

Route tables are... a table of rules, called routes, that decide where the data in your network should go.

Routes tables are needed to make a subnet public because... the table tells your subnet's traffic where to travel to send and receive data.

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Route destination and target

Routes are defined by their destination and target, which mean...the traffic is routed to the internet via the Internet Gateway.

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of...0.0.0.0 and a target of...internet gateway.

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Security groups

Security groups are...virtual firewalls controlling inbound and outbound traffic for cloud resources like ec2 instances, mainly in AWS using rules for network access management.

Inbound vs Outbound rules

Inbound rules are...specific configuration within a security group that defines what type of incoming network traffic is allowed to reach a resource. I configured an inbound rule that...it sends and receives data anywhere in a public network.

Outbound rules are...configurations within a security group that controls the network traffic allowed to leave a resource and travel to external destinations. By default, my security group's outbound rule...allows access to send data to an ip address whether its a VPC or other in a public network.

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Network ACLs

Network ACLs are...stateless rules at the subnet level that controls inbound and outbound traffic, unlike security groups which are stateful.they evaluate rules in order and can allow or deny specific traffic in aws.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that...a network acl is your second layer of defense at the subnet level while a security group is your first layer of defense at the instance level.

Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will...allow access to the internet for the nextwork vpc on the public network.

In contrast, a custom ACL's inbound and outbound rules are automatically set to... default it is designed to allow all traffic to move freely until you decide to customize the rules to fit your needs.

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